Mathematical Addendum: Response to Critique on the Feasible Region Ω and Stochastic Dynamics

VSF Research Group

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Context

We thank the reviewer for identifying three structural gaps in the original existence and invariance argument for the feasible region $\Omega \subset \mathbb{R}^{128}$ under stochastic updates $x(t{+}1) = f_0(x(t)){+}\sigma\eta(t)$. All three points have been formally resolved and integrated into the revised manuscript (Sec. 3–4). Below is the consolidated mathematical statement.

1. Convexity of Ω

The global region $\Omega = \bigcap_i \{x : A_i(x)\}$ is not convex: the entropy bound $H(x) \le H_{\max}$ is an upper level set of a concave function (non-convex in general), and the ethical exclusion A_0 is a non-geometric predicate (covenant-hash validity) that need not preserve convexity. A direct application of Brouwer's fixed-point theorem on Ω is therefore unjustified.

Resolution. The reviewer is right — and the corrected argument no longer requires convexity at any step. It has three independent legs:

1. **Existence of an invariant measure (Krylov–Bogolyubov).** The state space is a compact set and the update kernel is Feller (f_0 continuous, additive noise), so an invariant probability measure exists (Kryloff & Bogoliouboff [1937]). Compactness suffices; convexity plays no role.

2. **Deterministic fixed point (Banach).** On the contractive core

$$\Omega_A = \big\{x \in \Omega : \langle x, s\rangle \ge \theta_A \|x\|\,\|s\|\big\}, \qquad \theta_A = 0.88,$$

   — the intersection of a circular cone (convex for $\theta_A \ge 0$) with the bounded state set, nested inside the viable region $\Omega \supseteq \{\cos \ge 0.70\}$ — the measured contraction $\lambda_{\max} = \ln(0.99) < 0$ (EXP-LY-01), an average per-step log-contraction along core trajectories, makes some iterate f_0^n a contraction; the Banach principle applied to f_0^n then gives a unique fixed point of f_0. The argument requires completeness and invariance of the core, but not convexity.

3. **Uniqueness and rate (Foster–Lyapunov).** Uniqueness of the invariant measure and geometric convergence follow from the drift condition of Sec. 3 below (Meyn & Tweedie [2009]).

The sum-form route via Krasnosel'skiĭ's theorem for a contraction plus a compact continuous perturbation (Krasnosel'skiĭ [1955]) remains available, but it is no longer load-bearing: no step of the corrected argument depends on the convexity of Ω or of Ω_A.

2. Continuity and Invariance of f_0

Forcing $f_0(\Omega) \subseteq \Omega$ by projection would distort the natural dynamics and obscure failure modes. We do not impose invariance globally.

Resolution. The deterministic map f_0 exhibits layered behavior, and we are explicit about what is measured and what is not claimed:

- On Ω_A: the measured $\lambda_{\max} = \ln(0.99) < 0$ certifies average per-step log-contraction along core trajectories (equivalently, some iterate of f_0 is a contraction — the form used in Sec. 1). The global worst-case Lipschitz bound computable from the deployed constants, $L_f \leq \max_s \|\nabla \ell_s\| \cdot \operatorname{diam}\{\mu_s\}/\tau$, is not informative on the core: there the softmax mixture is near-saturated and its local gradient lies far below the worst case.
- On $\Omega \setminus \Omega_A$: we do *not* claim per-step non-expansiveness. The same worst-case bound evaluates to $L_f \in [1.4,\ 24.4]$ across the deployed temperature range, so a per-step bound $L \leq 1$ cannot be discharged from the code constants. Confinement off the core is carried empirically instead: excursions recover within a measured finite time τ_{rec} (sub-30 s across 20 forced thermal cycles), and sustained noise produces zero exits from Ω (Sec. 3).

No artificial clamping is applied. The geometry of the Lyapunov function $L(x)$ ensures that states perturbed outside Ω_A are naturally re-contracted into the core without leaving Ω (Remark 2, Sec. 3). Invariance is thus a *measured dynamical property*, not an imposed algebraic constraint.

3. Stochastic Perturbations $\sigma \cdot \eta(t)$

Even if $f_0 : \Omega \to \Omega$ were invariant, the additive noise $\sigma\eta(t)$ with unbounded support breaks pointwise invariance: $x(t+1) \notin \Omega$ with non-zero probability.

Resolution. Two observations. First, in the deployed system the perturbation is bounded by construction: the state lives on a compact set and the noise amplitude is governed by the A_{breath} axiom, so the unbounded-support premise applies to the idealized model, not to the running kernel. Second, for the idealized model we replace pointwise invariance with *stochastic viability* (Aubin [1991]) and the Lyapunov drift condition

$$\mathbb{E}\big[L(x(t+1)) - L(x(t)) \mid x(t) \in \Omega\big] \leq 0,$$

which is exactly the geometric-ergodicity condition for the induced Markov operator (Meyn & Tweedie [2009]). Confinement is then measured: EXP-NS-01 drives the kernel with sustained bounded noise at every tick and observes *zero exit events* over $N = 3000$ ticks per noise level (empirical exit rate $< 3/N$ at 95% confidence) for all $\sigma \in [0,\ 4.0\,\|x\|]$. The viability margin $m(\sigma)$ degrades smoothly and saturates at ≈ 0.33; no finite noise ceiling was observed. The system does not rely on hard boundaries; it relies on negative expected drift that recontracts excursions in finite time.

Integration with the Manuscript

The three objections are resolved in the current revision:

1. **Sec. 3, Def. 2 & Thm. 1:** Existence carried by Krylov–Bogolyubov (compactness + Feller) and the Banach principle on the core; convexity no longer enters the argument at any point.

2. **Sec. 4, Thm. 2 & EXP-LY-01:** Convergence in expectation via measured Lyapunov drift ($\lambda_{\max} = -0.010050$), not assumed invariance.

3. **Sec. 6.4, EXP-NS-01:** Sustained-noise margin curve replaces the borrowed impulsive proxy; zero observed exits up to $\sigma = 4.0\,\|x\|$ verify stochastic viability empirically.

The existence proof is now *measured, not postulated*: contraction, confinement, and ergodicity are discharged by hardware experiments rather than asserted algebraically. Brouwer's theorem is no longer invoked; the negative Lyapunov exponent and the viability-theoretic reading of Ω carry the argument.

References

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[2] M. A. Krasnosel'skiĭ, "Two remarks on the method of successive approximations," *Uspekhi Mat. Nauk*, 10(1):123–127, 1955.

[3] N. Kryloff and N. Bogoliouboff, "La théorie générale de la mesure dans son application à l'étude des systèmes dynamiques de la mécanique non linéaire," *Ann. of Math.*, 38(1):65–113, 1937.

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